

SUMIT JAHAGIRDAR

Lead Data Scientist | Applied AI Engineer | LLM Systems | Multimodal ML | Databricks
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PROFESSIONAL SUMMARY

Lead Data Scientist with 4+ years building production AI systems across enterprise analytics, document intelligence, computer vision, code intelligence, and generative media. Owns problem scoping, PRDs, architecture, implementation, deployment, and evaluation. Strong in Python, LLM pipelines, agent runtimes, multimodal extraction, semantic retrieval, Databricks, MLflow, FastAPI, and production ML infrastructure.

CORE SKILLS

AI / ML: LLMs, semantic search, NLP, computer vision, ASR, multimodal extraction, prompt optimization, contrastive learning, MLflow
Engineering: Python, PyTorch, FastAPI, Docker, Kubernetes, AWS Lambda, MCP, CI/CD, API design, backend architecture

EXPERIENCE

Lead Data Scientist | Aidetic | Bengaluru, India | Jun 2022 - Present

- Architected and shipped production AI systems across enterprise analytics, legal tech, financial parsing, call QA, computer vision, and generative media.
- Built Agent Factory, a custom state-separated agent runtime for multilingual enterprise analytics, including conversation state, execution state, semantic state, data objects, ACL-aware tool access, lineage, metadata registries, and evaluation hooks.
- Designed backend architecture, API surfaces, UX flows, sprint topology, CI/CD paths, and production deployment plans for client-facing AI systems.
- Developed Python execution engines for agents to run governed statistical computation and data analysis against enterprise data sources.
- Built AI Call Analyzer: Whisper ASR plus LLM grading rubrics for automated QA across compliance, objection handling, and script adherence; optimized batch inference through vLLM and prompt redundancy pruning.
- Built legal and financial document parsers for 200+ page credit agreements, SEC-style master tables, chart/table extraction, clause normalization, and cross-reference tracing.
- Delivered computer vision systems for ball tracking, vehicle damage detection, chart reconstruction, and object-level document understanding using OpenCV, PyTorch, YOLO, LayoutLM, and custom detectors.
- Built semantic retrieval and code intelligence systems using domain-tuned embeddings, vector search, semantic graphs, PageRank-based context ranking, and MCP integration for coding IDEs.

Software Engineer, Edison AI Team | GE Healthcare | Bengaluru, India | Aug 2021 - May 2022

- Engineered scalable microservices and Python SDK components for clinical ML workflows running on AWS Lambda and Kubernetes.
- Built reusable interfaces for model execution, workflow orchestration, and application integration inside Edison AI platform services.
- Worked on production-grade healthcare AI infrastructure with emphasis on reliability, packaging, deployment, and developer usability.

Machine Learning Research Intern | NVIDIA | Remote / India | Apr 2019 - Apr 2020

- Applied ML research in medical imaging, focused on few-shot learning and unsupervised clustering for pathology detection.
- Built experimental training pipelines, evaluation scripts, and model analysis utilities for computer vision research workflows.

SELECTED SYSTEMS AND PROJECTS

Agent Factory - Enterprise Agent Runtime | LLMs, Databricks, FastAPI, state management, ACL, lineage

- Built a multilingual zero-SQL analytics runtime for enterprise users, isolating conversation, execution, semantic, and data states for safer agent behavior.
- Implemented governed tool access, deterministic lineage, metadata registries, embedding-backed context retrieval, and evaluation hooks for auditable responses.

Credit Agreement Parser | Python, LLMs, NLP, document graphs, OCR, semantic search

- Parsed long-form credit agreements into structured financial/legal tables using segmentation, clause normalization, nested-definition handling, and cross-reference tracing.

AI Call Analyzer | Whisper, vLLM, LLM grading, batch inference, Python

- Converted audio calls into rubric-based QA scores for compliance, objection handling, and script adherence; optimized throughput through vLLM batch execution.

Generative AI Video Editor | Whisper, Coqui TTS, MakeltTalk, audio/video processing

- Built a transcript-driven video editor where text edits produce timestamped cuts, cloned replacement speech, localized lip-sync patches, and reconstructed MP4 outputs.

Computer Vision Systems | OpenCV, PyTorch, YOLO, LayoutLM, object detection

- Delivered cricket ball tracking, vehicle damage detection, chart reconstruction, and document layout extraction pipelines

EDUCATION

B.Tech, Electronics and Communication Engineering | VIT Vellore | 2017 - 2021 | CGPA: 8.03 | IoT and Sensors specialization
Technical Head, IET VIT; Microsoft Student Ambassador (Gold)

RECOGNITION

NVIDIA GTC 2020 Speaker; Aidetic Launchpad Winner; GE Precision Health Winner; IET PATW 2020 Winner; Top 30 Microsoft Student Partners APAC.